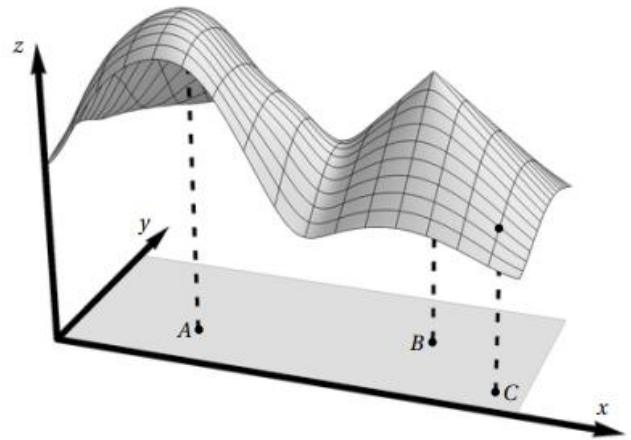


3.4 Differentiability Worksheet

1. Consider the function $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ whose graph is shown at the right. Let A and B be points in $\mathbb{R}^2$ corresponding to two “peaks” of the graph, and C be the point in $\mathbb{R}^2$ corresponding to the dot on the graph. For each part, circle the answer that is the most consistent with the picture.



- a) At the point A , the function g is:

continuous differentiable **both** neither

- b) At the point B , the function g is:

continuous differentiable both neither

- c) At the point C , the function $\frac{\partial g}{\partial x}$ is:

negative zero positive

2. Show that $f(x, y) = xe^{xy}$ is differentiable at $(1, 0)$.

We see that $f_x(x, y) = xe^{xy}y + e^{xy} = e^{xy}(xy + 1)$ and $f_y(x, y) = x^2e^{xy}$.

We see that both functions are clearly continuous at all points therefore continuous at $(1, 0)$.

Therefore, we can conclude that $f(x, y)$ is differentiable at the point $(1, 0)$.

3. Show that $f(x, y) = 1 + x \ln(xy - 5)$ is differentiable at $(2, 3)$

We see that $f_x(x, y) = 0 + x \left(\frac{1}{xy - 5} \right) (y) = \frac{xy}{xy - 5}$ and $f_y(x, y) = 0 + \frac{x}{xy - 5} (x) = \frac{x^2}{xy - 5}$.

We see that both functions are not always continuous but are continuous at $(2, 3)$.

Therefore, we can conclude that $f(x, y)$ is differentiable at the point $(2, 3)$.

4. Consider the function $f(x, y) = \frac{x^{12} + y^4 + 36}{3x - 1}$.

a) Confirm that the function $f(x, y)$ is differentiable at the point $(1, 1)$.

$$\text{We see that } f_x(x, y) = \frac{(3x-1)(12x^{11}) - 3(x^{12} + y^4 + 36)}{(3x-1)^2} \text{ and } f_y(x, y) = \frac{4y^3}{3x-1}.$$

We see that both functions are rational functions which clearly exist and are continuous at $(1, 1)$.

Therefore, we can conclude that $f(x, y)$ is differentiable at the point $(1, 1)$.

b) Confirm that the function $f(x, y)$ is continuous at the point $(1, 1)$.

From part (a) we know that $f(x, y)$ is differentiable at the point $(1, 1)$. This automatically implies that $f(x, y)$ is continuous at the point $(1, 1)$.

Note: you could also check the continuity of $f(x, y)$ at $(1, 1)$ by using the definition of continuity, however this approach will take more work than using the important fact used above from section 3.4.

5. Consider the function $f(x, y) = \sqrt{x + e^{4y}}$.

a) Confirm that the function $f(x, y)$ is differentiable at $(3, 0)$.

$$\text{We see that } f_x(x, y) = \frac{1}{2}(x + e^{4y})^{-1/2} (1) = \frac{1}{2\sqrt{x + e^{4y}}} \text{ and } f_y(x, y) = \frac{1}{2}(x + e^{4y})^{-1/2} (4e^{4y}) = \frac{2e^{4y}}{\sqrt{x + e^{4y}}}.$$

We see that both functions are not always continuous but are continuous at $(3, 0)$.

Therefore, we can conclude that $f(x, y)$ is differentiable at the point $(3, 0)$.

b) Confirm that the function $f(x, y)$ is continuous at the point $(3, 0)$.

From part (a) we know that $f(x, y)$ is differentiable at the point $(3, 0)$. This automatically implies that $f(x, y)$ is continuous at the point $(3, 0)$.

Note: you could also check the continuity of $f(x, y)$ at $(3, 0)$ by using the definition of continuity, however this approach will take more work than using the important fact used above from section 3.4.